

3D Coordinates Extension 2 — English Worksheet (Beginner)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Beginner (Extension, after Lesson 3) ·

Time: about 30 minutes

Build a mob head in 3D. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

 Stand on your **home spot** (red block) every run. Move, and the head moves too.

1 · Predict

Read the code. Circle one word.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. Circle one: across · up · deeper

```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Only y changes. Circle one: across · up · deeper

2 · Spot the Bug 🐛

This eye should be one step **deeper**, at (4, 7, 2). The code below only has **two** numbers, so it lands on the front.

```
# broken  
place black block at (4, 7)
```

Fill the blank to fix it:

```
place black block at (4, 7, __)
```

Which number fills the blank? Circle one: 2 · 1

What was missing? Circle one: the z number · the color

3 · Show Your Work 📷 🎥

Look at the 🟢 creeper head in the world. Walk around it.

Build a small solid head — a cube. Keep it easy:

Front face: fill green blocks from (3, 5, 1) up to (5, 7, 1) — a small square.

Add depth: build the same square again at $z = 2$, then $z = 3$.

Add 2 eyes on the front only: black block at (3, 7, 1) and black block at (5, 7, 1).

💡 Tip: the eyes only show on the **front**. The sides and back are plain green.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.